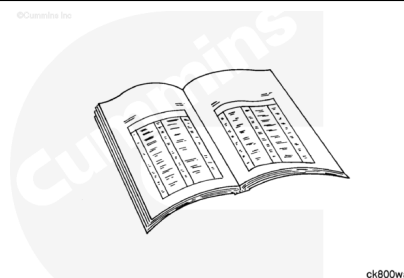


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

- Remove the hydraulic pump. Refer to the OEM troubleshooting and repair manual.

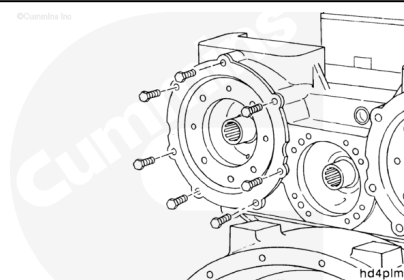
Remove

Remove the outer hydraulic pump support drive.

Note : Some engines will not have an outer hydraulic drive. These engines require a cover plate and a non-splined shaft, but do not use a hydraulic gear.

If using a cover plate, remove the cover plate and non-splined shaft.

Remove the seven capscrews from each outboard hydraulic drive or cover plate.



⚠ WARNING ⚠

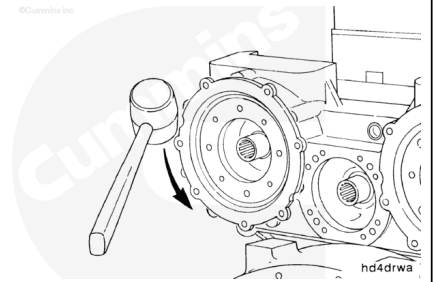
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this

component or assembly.

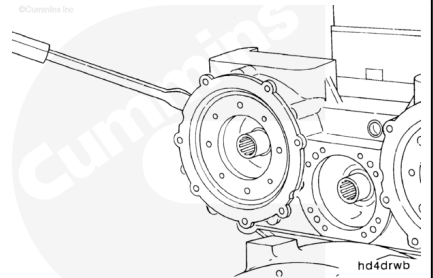
⚠ CAUTION ⚠

Remove the outboard hydraulic pump supports as described below to avoid breaking the capscrew mounting flanges.

The outboard hydraulic pump supports **must** be rotated to be removed from the housing. Use a mallet. Carefully tap the side of the support.



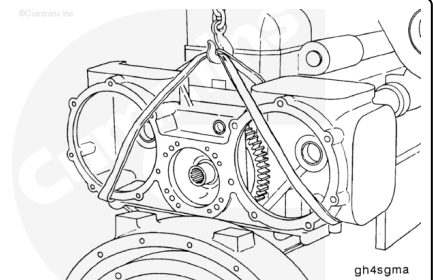
Use a pry bar. Remove both outboard hydraulic pump supports. Remove and discard the gaskets.



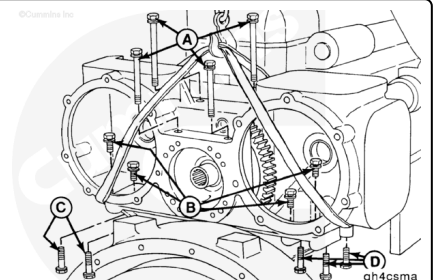
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Use a hoist and lifting sling. Install the lifting sling around both sides of the upper housing. Adjust the hoist until there is tension in the lifting sling.

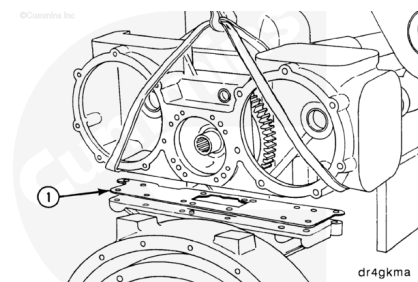


Remove the four capscrews from the top of the housing (A), the four internal capscrews (B), the two capscrews from the bottom of the upper housing on the left side (C), and the three capscrews from the bottom on the right side (D).



⚠ CAUTION ⚠

The hydraulic gear protrudes out of the bottom of the housing. Put the upper housing upside down on a workbench to prevent damage to the gear or bushing.

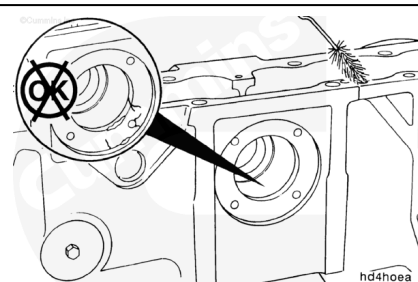


Remove the rear gear train upper housing from the lower housing. Remove the gasket (1). Keep the gasket for future use.

Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

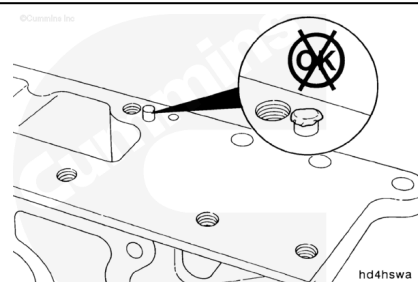


Use a bore cleaning brush and safety solvent. Clean and inspect the upper housing for damage. Clean and inspect all oil drillings for debris.

Note : Do not remove the 1-inch internal hex pipe plug in the upper housing near the center drive bore. Do not remove the 1/8-inch pipe plug that is on the same housings located on the engine side of the housing into the right side outboard hydraulic bushing bore.

Inspect the two rear gear train upper housing dowel pins in the top of the lower housing. Any damaged dowels **must** be replaced.

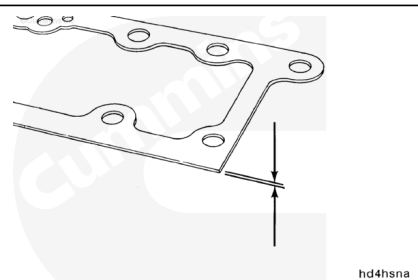
Use dowel pin remover, Part Number 3163720, to remove the dowel pins. Use a brass or lead hammer to install the dowel pins.



Measure the thickness of the gasket used on the rear gear train upper housing.

Refer to the following table for the part number and dimension.

	mm		in

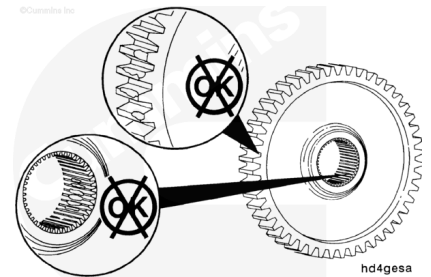


Part Number 205723	0.38	NOM	0.015
Part Number 3077231	0.51	NOM	0.020
Part Number 3201852	0.76	NOM	0.030

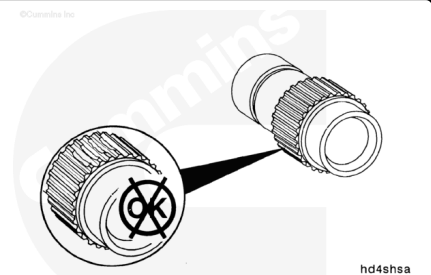
Note : Used gaskets measure thinner.

Clean and check the center gear for damage.

Check the gear teeth for excessive fretting. Check the internal splines for wear.



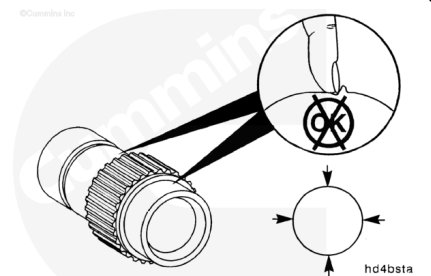
Check the center drive shaft. Check the external splines for wear.



Inspect the bushing diameters on the shaft for wear. If a fingernail will catch on scratches or grooves, replace the shaft.

Measure the shaft of the bushing diameters.

mm		in
47.536	MIN	1.8715
47.549	MAX	1.8720

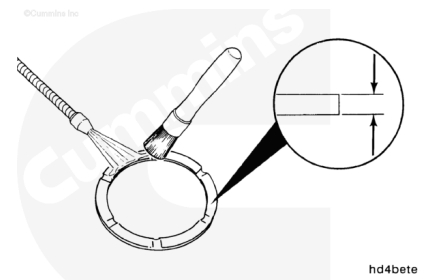


If the bushing does **not** meet the specifications, the bushing **must** be replaced.

Inspect the thrust bearings.

Measure the thrust bearing thickness.

mm		in
2.27	MIN	0.085
2.31	MAX	0.091



If the thrust bearings do **not** meet the specifications, the bearings **must** be replaced.

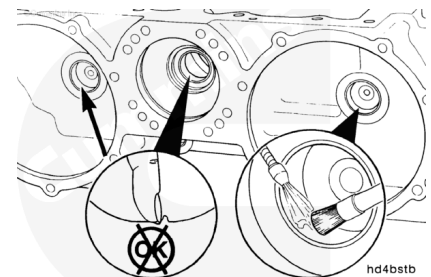
Check the retaining rings for damage.

Check the spacer bearing for damage.

Inspect the three hydraulic pump support bushings in the upper housing. If a fingernail will catch on scratches or grooves, replace the bushing.

Measure the bushing inside diameter.

mm		in
47.60	MIN	1.874
47.68	MAX	1.877

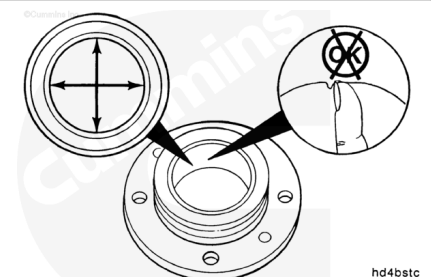


If the bushing does **not** meet the specifications, the bushing **must** be replaced.

Inspect the hydraulic pump support bushing in the shaft support plate. If a fingernail will catch on scratches or grooves, replace the bushing.

Measure the bushing inside diameter.

mm		in
47.60	MIN	1.874
47.68	MAX	1.877

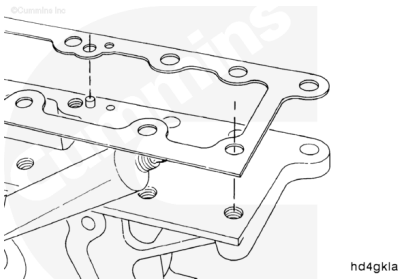


If the bushing does **not** meet the specifications, the bushing **must** be replaced.

Install

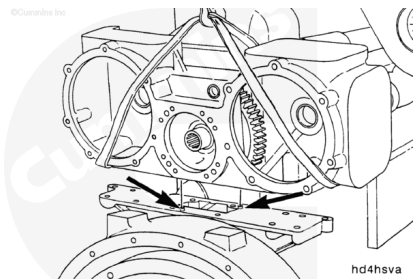
Use gasket adhesive on the gasket. Do **not** use an excessive amount of adhesive on the gasket.

Align and install the rear gear train upper housing gasket to the rear gear train lower housing.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

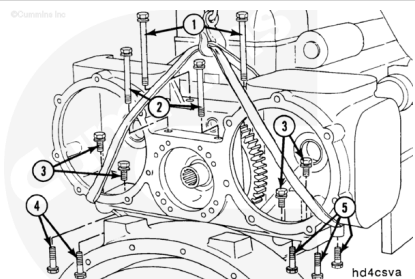


⚠ CAUTION ⚠

Use extra care when aligning the upper housing onto the lower housing to prevent gasket damage.

Use the dowel pins to align the upper housing onto the lower housing.

Use **only** SAE Grade 8 capscrews to install the upper housing. Install the following capscrews according to the table below.

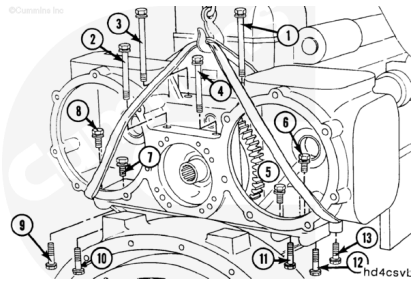


Ref	Location	Part No.	Size [Inch]	Quantity
1	Top-Front	3202224	1/2-13 x 9 1/4	2
2	Top-Rear	3202223	1/2-13 x 7 1/2	2
3	Internal	169195	1/2 -13 x 1 1/4	4
4	Bottom-Left Side	3202222	1/2-13 x 1 3/4	2
5	Bottom-Right Side	3202222	1/2-13 x 1 3/4	3

Use the sequence shown to tighten the capscrews.

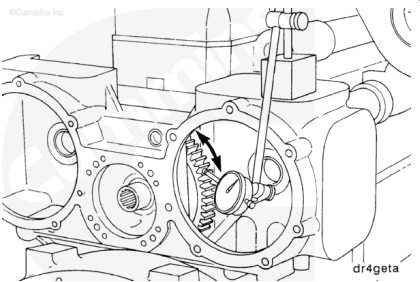
Torque Value:	Step 1	70 n.m [50 ft-lb]
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	Step 2	150 n.m [110 ft-lb]
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Use a dial indicator. Be sure the lower idler gear is secure. If the gear is **not** secure, the indicator reading includes the lower idler gear to the crankshaft gear backlash.

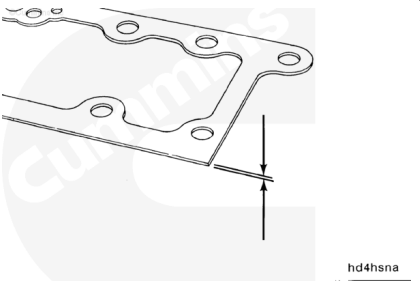
Measure the backlash between the center drive gear and the lower idler gear.



mm		in
0.05	MIN	0.002
0.51	MAX	0.020

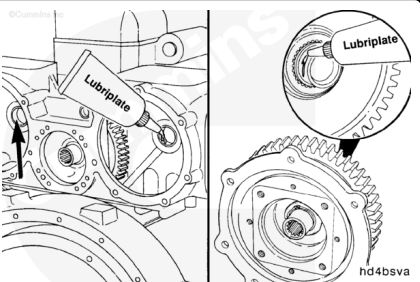
If the gear lash is less than the specified limits, thicker gaskets are available. Do **not** use more than two gaskets to correct the backlash.

If the gear lash is greater than the specification and the thinnest gasket is used, the upper and lower gears **must** be replaced.



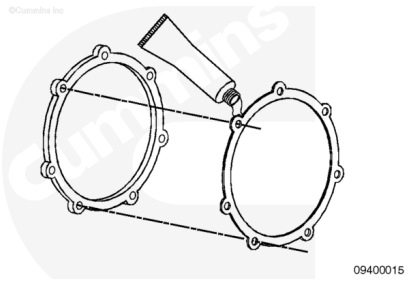
Install the outer hydraulic pump support drive.

Use Lubriplate® 105, or equivalent. Lubricate the outer hydraulic pump support bushings in the upper housing. Lubricate the bushing surfaces on both shafts in the outer hydraulic pump support drive assemblies.



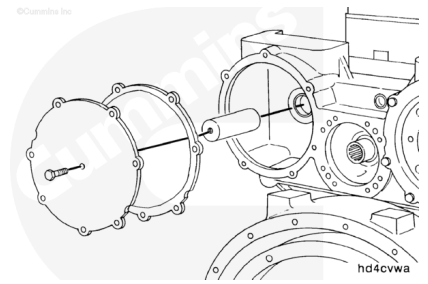
Note : Some engines do not have an outer hydraulic pump support drive. These engines require a cover plate and a non-splined shaft, but do **not** use a hydraulic gear.

Use gasket adhesive sparingly. Install the hydraulic support housing gaskets on both support mounting flanges on the outer hydraulic pump support drive assemblies.

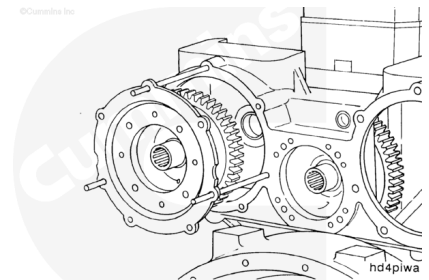


If using the outer hydraulic pump support cover plate, install the non-splined shaft onto the cover plate. Use a [3/8-16 x 1-1/4 in] capscrew.

Torque Value: 40 n•m [30 ft-lb]

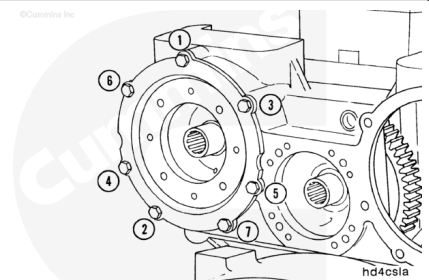


Use three [7/16-14 x 4 inch] guide studs. Install one outer hydraulic pump support drive assembly.



Install four capscrews. Hand-tighten the capscrews. Remove the guide studs. Install the remaining three capscrews. Use the sequence and apply torque as shown.

Torque Value: 70 n•m [50 ft-lb]



Finishing Steps

- Install the hydraulic pump. Refer to the OEM troubleshooting and repair manual.
- Operate the engine and check for leaks.



